

Experiment 18

Prolog Program for STUDENT-TEACHER-SUB-CODE

1. Aim

To write and execute a Prolog program to create a database linking students, teachers, and subject codes, and to retrieve which teacher teaches a given student using facts, rules, and queries.

2. Objective

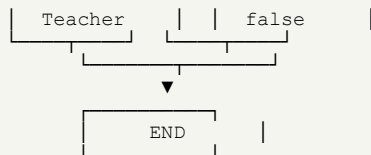
- To understand database representation in Prolog using multiple related facts.
- To learn how to represent students, teachers, and subject codes as facts.
- To retrieve information by joining facts across predicates using a rule.
- To understand Prolog's pattern matching and logical search across relations.

3. Algorithm

1. Start the program.
2. Create a Prolog database using the predicates student/2, teaches/2.
3. Store each student's name with the subject code they are enrolled in.
4. Store each teacher's name with the subject code they teach.
5. Define a rule find_teacher/2 to link a student to their teacher via the common subject code.
6. Enter the student's name as a query.
7. Search the database for a matching subject code and teacher.
8. If found, display the teacher's name; otherwise Prolog returns false.
9. Stop the program.

4. Flowchart





5. Code with Output

Code:

```

% -----
% STUDENT - TEACHER - SUB_CODE Database
% -----

% teaches(Teacher, SubCode).
teaches(mrs_shah, cs101).
teaches(mrs_verma, cs102).
teaches(mr_gupta, cs103).

% student(StudentName, SubCode).
student(arjun, cs101).
student(bhavna, cs102).
student(chirag, cs103).

% -----
% Rule to find teacher of a student
% -----
find_teacher(Student, Teacher) :-
    student(Student, Code),
    teaches(Teacher, Code), !.

find_teacher(_, not_found).

```

Output:

```

?- find_teacher(arjun, T).
T = mrs_shah.

?- find_teacher(bhavna, T).
T = mrs_verma.

?- find_teacher(rohit, T).
T = not_found.

```

6. Result

The Prolog program was successfully implemented to link students to their subject teachers via a common subject code, using facts and a join rule.

7. Conclusion

Thus, the STUDENT-TEACHER-SUB-CODE database was successfully implemented in Prolog. The experiment demonstrates how facts from two related predicates can be joined through a shared field (subject code) to answer relational queries.